

Section 1108's Long Shadow: Sixty Years of Capped Medicaid in the U.S. Territories, 1968–2024

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Running title

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Conflicts of interest

The author declares no financial, commercial, or personal conflicts of interest related to this work.

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Data availability statement

All primary data sources are public. The CDC WONDER Compressed Mortality and Underlying Cause of Death systems, the CDC WONDER Natality system, the ICPSR Multiple Cause of Death public-use files, the NCHS Vital Statistics of the United States Volume I, IPUMS USA and the Puerto Rico Community Survey, the MACPAC *Medicaid and CHIP in the Territories* report series, the CMS Financial Management

Report annual workbooks, the HCFA Statistical Supplement and HCFA Data Compendium series, and the Puerto Rico Demographic Registry's *Informe Anual de Estadísticas Vitales* are all retrievable from public archives. Source-provenance sidecars for every non-trivial raw file are deposited in the project repository.

Replication code, the machine-extracted Puerto Rico Demographic Registry corpus (559 validated outcome cells), the donor-state pre-1968 IMR panel (126 cells), the state-year CDC WONDER Natality births panel (1,530 cells), analytic panels for all four arms, and the manuscript source are deposited at <https://github.com/jonpalisoc1024/papers> (territorial-medicaid-caps subdirectory). The Puerto Rico Demographic Registry *Informe Anual de Estadísticas Vitales* corpus is released as a reusable artifact for future Puerto Rico health-policy research.

Author contributions

Jonathan Palisoc is the sole author. He conceptualized the study, assembled the data, developed the Puerto Rico Demographic Registry corpus extraction code, conducted all analyses, drafted the manuscript, and is responsible for the final version.

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are my own.

Policy Points

- Congress should enact a successor to the Consolidated Appropriations Act of 2022 territorial Medicaid supplements before their FY2027 expiration; lapse would re-impose a roughly 95 percent cap-bite against Puerto Rico’s state-equivalent need.
- The Medicaid and CHIP Payment and Access Commission should commission a post-unwinding successor analysis extended to the four smaller territories, whose vital-statistics archives this analysis could not reach.
- The Congressional Budget Office and Joint Committee on Taxation should score state per-capita-cap proposals against two infant-mortality anchors (Currie–Gruber; Goodman-Bacon) implying approximately 4,500 to 14,800 excess infant deaths per year under a 1968-scaled cap.

Abstract

Context. Section 1108(c) has capped federal Medicaid contributions to the U.S. territories since 1968 — the same per-capita-block-grant architecture recent state restructuring proposals would impose nationally. The historical consequences have not been quantified with modern causal-inference designs, partly because Puerto Rico vital-statistics data sit in a Spanish-language Demographic Registry corpus standard federal systems do not carry.

Methods. We assemble a 1950–2024 panel of territorial and state Medicaid financing, coverage, and population-health outcomes, anchored by a machine-extracted corpus of the Puerto Rico Demographic Registry’s *Informe Anual de Estadísticas Vitales* (559 validated cells; thirteen outcomes). We evaluate four cap-regime shifts: the 1968 Section 1108(c) imposition, the 2011 ACA Section 2005 partial uncapping, and

the FY2020 cliff (each with synthetic control), plus a reverse counterfactual applying published Medicaid-mortality elasticities to the fifty-state panel under three hypothetical per-capita-cap regimes.

Findings. The 1968 imposition is not associated with a detectable infant-mortality penalty ($ATT = +0.08$ per 1,000; placebo $p = 0.20$). The 2011 partial uncapping is not associated with a detectable coverage change ($ATT = -0.26$ percentage points; $p = 0.93$). The FY2020 cliff is confounded by the COVID-19 public health emergency and bridge supplements; Puerto Rico’s federal Medicaid per capita grew 52 percent FY2019–FY2024 against ≈ 70 percent donor growth. A 1968-scaled state cap would remove \$513 billion in FY2024 and, under two infant-mortality anchors (Currie–Gruber; Goodman-Bacon), imply $\approx 4,500$ – $14,800$ excess infant deaths per year.

Conclusions. The historical cap regime shows no detectable infant-mortality differential, most plausibly because of gradual tightening, floor compression, and rare-event noise rather than the absence of a cap-on-mortality elasticity. The policy-relevant magnitudes lie in the prospective fiscal scale of state-level proposals and the literature-anchored mortality range those caps imply. The case for territorial equity rests on the fiscal-architecture imposition, the structural disadvantage of the cap-bound population, and the state-level cap magnitudes visible in the reverse-counterfactual grid.

1. Introduction

The federal contribution to Medicaid in the fifty states and the District of Columbia is open-ended. The federal medical assistance percentage moves with each state’s per-capita income, is rebated against actual qualifying expenditure, and carries no annual ceiling. This open-ended-entitlement architecture is the foundation of Medicaid’s countercyclical role in state budgets and of its expansion under the Affordable Care Act.^{1,2}

The federal contribution to Medicaid in the five U.S. territories — the Commonwealth of Puerto Rico, the

U.S. Virgin Islands, Guam, American Samoa, and the Commonwealth of the Northern Mariana Islands — has, since 1968, been capped under Section 1108(c) of the Social Security Act. Each territory receives an annual block grant set in nominal dollars, which the territory matches at a statutorily fixed federal share that has historically been far below the share to which the territories’ per-capita-income statistics would otherwise entitle them.^{3–5} When a territory’s Medicaid program exceeds the cap or exhausts the matching draw, the federal contribution stops. The territorial Medicaid program is, structurally, a per-capita-capped block grant — the same architecture that the Graham–Cassidy proposals of 2017 and a series of later proposals would have imposed on state Medicaid programs.^{6,7} The per-capita-cap logic has also re-entered the national debate alongside the work-requirement provisions of the One Big Beautiful Bill Act: both rest on a shared fiscal-restraint posture toward Medicaid that the territorial regime has tested at scale for sixty years.

This architectural difference matters in two distinct ways. The first is normative: the residents of all five territories are U.S. citizens or U.S. nationals, and an unequal federal contribution to the medical care of U.S. citizens raises questions for political, legal, and ethical scholarship that this paper does not principally address.⁸ The second is empirical: if the cap regime imposes adverse mortality, coverage, or care-access consequences on Puerto Rico that would be observable in the historical record, then the prospective imposition of a similar cap on state Medicaid would be expected to produce similar consequences at scale. If the cap regime does not impose detectable adverse consequences in the Puerto Rico historical record, the policy debate is shifted: the case against state-level per-capita caps must be made primarily through fiscal-architecture and prospective-elasticity reasoning rather than through historical analogy.

The empirical literature on this question is thin. Descriptive analyses by the Medicaid and CHIP Payment and Access Commission, the Government Accountability Office, the Kaiser Family Foundation, and the Congressional Research Service document the financing differential and its history.^{3–5,9} A causal-inference literature on the territories’ health outcomes does not exist at scale. Portela and Sommers compared insurance and access to care between Puerto Rico and selected mainland populations using descriptive statistics,⁸ Perreira,

Jones, and Oberlander analyzed the post-2014 Puerto Rico Medicaid funding crisis as policy commentary,¹⁰ and Kishore et al., Santos-Lozada and Howard, and Roman examined post-Maria mortality.^{11–13} None exploits the 1968 §1108(c) imposition, the 2011 partial uncapping, or the FY2020 cliff in a synthetic-control or other quasi-experimental design.

The principal obstacle has been data. The Centers for Disease Control and Prevention’s WONDER mortality system does not include Puerto Rico in any vintage. The National Center for Health Statistics’ Multiple Cause of Death microdata captures Puerto Rico residents only when they die on the U.S. mainland — typically six to fifty deaths per year, a migration measure rather than a territory-mortality measure. The Puerto Rico mortality and natality series for the cap-era window 1968–2008 reside almost exclusively in the Spanish-language *Informe Anual de Estadísticas Vitales* of the Puerto Rico Department of Health Demographic Registry — a sixty-year corpus of fifty-nine PDF volumes that has not, to our knowledge, been previously machine-extracted to a tidy panel for econometric analysis at this scale.

This paper has four objectives. First, we extract the Puerto Rico Demographic Registry corpus into a thirteen-outcome validated panel covering 1950–2023 and demonstrate that the resulting series support modern quasi-experimental designs. Second, we conduct three historical synthetic-control analyses of regime shifts in territorial Medicaid financing — the 1968 §1108(c) imposition, the 2011 ACA Section 2005 partial uncapping, and the FY2020 cliff — and document the identified effects or, more often, the identified absence of effects on infant mortality, Medicaid coverage, and federal Medicaid spending per capita. Third, we conduct a prospective reverse-counterfactual analysis applying published Medicaid-mortality elasticities^{14–16} to the fifty-state Medicaid panel under three hypothetical per-capita-cap regimes. Fourth, we draw implications for the territorial-Medicaid-equity debate that will recur as the Consolidated Appropriations Act of 2022 territorial appropriations expire in FY2027.⁹

2. Background: Section 1108(c) and the Territorial Medicaid Regime

Title XIX of the Social Security Act, enacted in 1965, established Medicaid as a federal-state program with open-ended federal matching at each state's per-capita-income-based federal medical assistance percentage. In January 1968, Public Law 90-248 (the Social Security Amendments of 1967) amended Section 1108 of the Act to impose annual caps on federal contributions to territorial Medicaid programs, statutorily fixed in nominal dollars and not indexed for inflation, population growth, or medical-cost growth.⁹ The initial caps were \$20 million for Puerto Rico, \$900,000 for Guam, and \$650,000 for the U.S. Virgin Islands. These caps were below the federal contribution that would have flowed under open-ended matching even at the territories' then-current program sizes; the caps were therefore binding from year one. The territorial federal share was statutorily fixed at 50 percent (later 55 percent), far below the approximately 83 percent share to which Puerto Rico's per-capita income would have entitled it under the standard formula.^{3,9} By the late 1990s, Puerto Rico's actual federal contribution per Medicaid beneficiary was approximately 15 percent of the federal contribution an equivalent state Medicaid beneficiary would have received.³

Three companion institutional changes shaped Puerto Rico's post-2006 Medicaid revenue capacity in ways that compound the Section 1108(c) cap. First, the phase-out of Internal Revenue Code Section 936 — the federal tax credit that anchored Puerto Rico's pharmaceutical and manufacturing sector — was enacted in 1996 (Public Law 104-188) and completed by 2006. Section 936's repeal eroded payroll, corporate-tax, and personal-income-tax bases on the island over a decade and removed approximately one-quarter of pre-2006 Puerto Rico gross domestic product, sharply constraining the territory's ability to draw down its federal Medicaid match.⁹ Second, the 2010 Patient Protection and Affordable Care Act selectively extended health insurance reform to the territories: the individual market reforms applied, but the open-ended state Medicaid expansion did not, and Puerto Rico remains statutorily excluded from the ACA Medicaid expansion mechanism that has driven post-2014 coverage gains on the mainland. Third, the 2016 Puerto Rico Oversight,

Management, and Economic Stability Act (PROMESA, PL 114-187) placed Puerto Rico's fiscal affairs under a federal oversight board with authority over the territory's budget, including Medicaid appropriations, and reduced the territorial government's discretion to redirect general-fund revenue into Medicaid match. Together, the §1108(c) cap, the §936 phase-out, the ACA Medicaid-expansion exclusion, and the PROMESA fiscal-oversight regime form an institutional triad that constrains both the supply of federal Medicaid dollars to Puerto Rico and the territory's capacity to substitute local revenue for federal shortfalls.

Section 2005 of the Patient Protection and Affordable Care Act of 2010 (ACA) appropriated \$7.3 billion in supplemental territorial Medicaid funding, available July 1, 2011 through September 30, 2019, of which \$6.3 billion was directed to Puerto Rico.^{3,9} The Section 2005 supplement operated as a multi-year drawdown rather than as a cap revision; Puerto Rico drew approximately \$800 million per year, exhausting the fund by FY2019. Between FY2017 and FY2019, a series of disaster-related and budget-act supplements provided additional one-time bridges in response to Hurricane Maria's September 2017 landfall.

The exhaustion of Section 2005 and the reversion of Puerto Rico's federal share to its statutory baseline at the start of FY2020 were widely characterized as a "Medicaid funding cliff."^{4,9,10} The Further Consolidated Appropriations Act of 2020 (CAA 2020, PL 116-94) provided a four-year bridge of approximately \$2.6 billion per year through FY2023 and raised the territorial federal share to 76 percent for the bridge period. The Families First Coronavirus Response Act (FFCRA) added a temporary federal-share enhancement during the COVID-19 public health emergency. The Consolidated Appropriations Act of 2022 (CAA 2022, PL 117-103) extended Puerto Rico's enhanced financing through FY2027 with statutory annual caps of \$3.275 billion for FY2023 and \$3.325 billion for FY2024.⁹ The fiscal cliff was therefore not allowed to fully bind in 2020 or thereafter; what the third historical arm observes is the joint effect of the statutory cliff, the bridge supplements, the COVID-19 public health emergency Medicaid continuous-enrollment requirements, and Hurricane Maria's continuing health-system effects.

Each cap-regime shift is, in principle, an experiment that identifies a different parameter of the cap-on-Medicaid relationship. The 1968 imposition tests whether the introduction of capped Medicaid to a previously open-ended-matching jurisdiction was associated with differential adverse outcomes. The 2011 partial uncapping tests whether a large bilateral reduction in cap-bite produced an effect in the opposite direction — a falsification of any 1968 finding through sign reversal. The FY2020 cliff tests whether a bilateral reimposition of cap-bite produced a differential adverse effect. A unified design that includes a prospective state-level reverse counterfactual provides the policy-relevant magnitudes that the historical Puerto Rico arms cannot.

The territorial cap-bite landscape is heterogeneous in absolute dollars but uniform in fiscal architecture. Table 1 reports the FY2024 statutory federal Medicaid allotment cap and the statutory federal share for each of the five territories, together with the cumulative trajectory of the allotment cap from FY2006 (the first year with comparable MACPAC-tabulated allotments for all five territories) through FY2024. Puerto Rico’s cap rose from \$219 million in FY2006 to \$3,173 million in FY2024 — a fourteen-fold nominal increase driven almost entirely by the ACA Section 2005 drawdown, the post-Maria disaster supplements, and the CAA 2020/2022 bridge. The four smaller territories experienced parallel but smaller-magnitude bridges: USVI and Guam each rose from approximately \$13 million in FY2006 to approximately \$139 million in FY2024, CNMI from \$5 million to \$66 million, and American Samoa from \$6 million to \$94 million. All five territories saw their statutory federal share rise from 50 percent (FY2006–FY2010) to 55 percent (FY2011–FY2018) and to 76 percent (FY2019 and forward, under CAA 2020/2022 and FFCRA enhancements). In every territory and every year since 1968, the cap has been below the open-ended-matching federal contribution that the territory’s enrolled population would have generated under standard FMAP rules; the bridging supplements raised the cap toward — but not to — state-equivalent federal need.

Table 1. Statutory federal Medicaid allotment cap (millions of nominal dollars) and statutory federal share for the five U.S. territories, FY2006–FY2024.

Territory	FY2011						
	FY2006		cap				
	cap		(M, multi—		FY2024	FMAP	FMAP
	(M) FY2010	cap(S)	Section2005	Cap(SM)	drawnceil2006	FY2011	2024(M)
Puerto Rico (PR)	219	263		6,300	800	3,173	50% 76%
U.S. Virgin Islands (USVI)	13	16		268	19	139	50% 76%
Guam (GU)	12	15		268	19	139	50% 76%
American Samoa (AS)	6	8		181	13	94	50% 76%
Northern Mariana Islands (CNMI)	5	6		109	8	66	50% 76%

Note. FY2011 caps include the ACA Section 2005 drawdown ceiling for each territory. FY2024 caps reflect the CAA 2022 statutory schedule. FMAP = statutory federal medical assistance percentage. Source: authors' tabulation from MACPAC *Medicaid and CHIP in the Territories* (February 2021), MACPAC *MACStats*

Medicaid and CHIP Data Book (December 2023), and Congressional Research Service Report R47601 (Mitchell 2023).

Figure 1 maps the prospective state-level cap-bite distribution that would obtain under three counterfactual per-capita-cap regimes, illustrating how the territorial fiscal architecture would scale to the fifty-state Medicaid program.

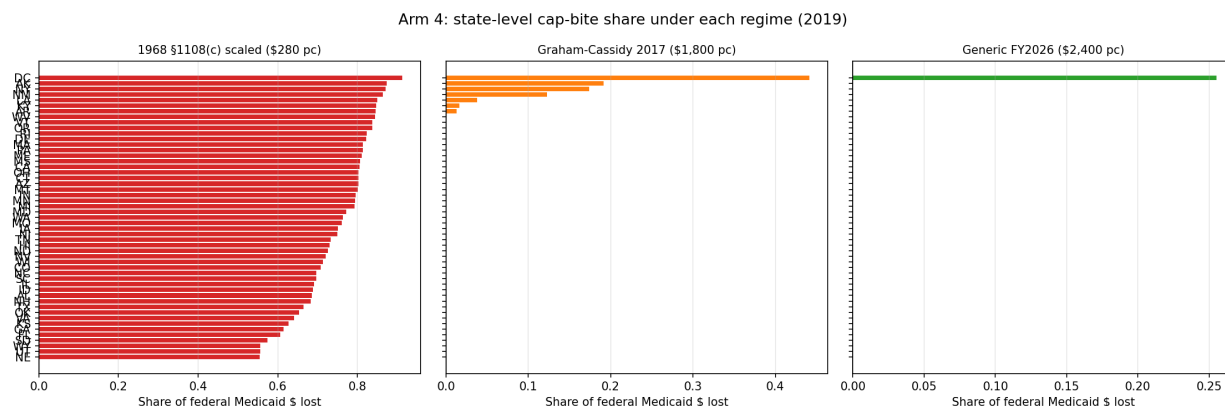


Figure 1: **Figure 1.** State-level federal Medicaid share that would be suppressed under three counterfactual per-capita-cap regimes, FY2024. Each panel shows the share of actual federal Medicaid spending that the cap would bind off. Under a 1968-scaled cap of \$280 per capita, every state and the District of Columbia would have its federal Medicaid contribution suppressed by the cap (the cap binds for all fifty-one jurisdictions). Under the Graham–Cassidy 2017 cap of \$1,800 per capita, the cap binds for 25 of 51 jurisdictions; under a generic FY2026-scale cap of \$2,400 per capita, the cap binds for 8 of 51. Source: authors’ calculations from MACPAC MACStats, CMS CMS-64 financial management reports, and Census population estimates.

3. Data

This study assembles a unified 1950–2024 panel of Puerto Rico, four-smaller-territory, and state Medicaid financing, coverage, and population-health outcomes. Treated units are the five U.S. territories; Puerto Rico is the primary treated unit, with the four smaller territories entering as secondary panels with substantially smaller cap-bite. The donor pool for the 1968 historical arm comprises ten Deep South states (Alabama, Arkansas, Georgia, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, West Virginia); the donor pool for post-2006 arms adds four high-Hispanic and Southwestern states (Ari-

zona, Florida, New Mexico, Texas) to better match Puerto Rico’s post-§936 demographic composition. The reverse-counterfactual arm uses the full fifty-state-plus-District-of-Columbia panel.

State-level mortality data come from CDC WONDER’s Compressed Mortality (1968–1998) and Underlying Cause of Death (1999–2024) systems and the Inter-university Consortium for Political and Social Research Multiple Cause of Death public-use files for individual-level pre-1979 ICD-8 detail.^{17,18} Pre-1968 donor-state infant deaths and live births are reconstructed from NCHS Multiple Cause of Death microdata aggregated to (state, year) and from NCHS *Vital Statistics of the United States* Volume I Table 2-1, transcribed cell-by-cell from the published volumes for fourteen donor states and nine years. State-year live births 1995–2024 for the reverse-counterfactual birth denominator come from CDC WONDER Natality interactive exports across 51 jurisdictions $\times$ 30 years; all four published-national-total validation anchors match at zero percent difference.

Insurance coverage outcomes come from the American Community Survey for the fifty states and the District of Columbia and from the Puerto Rico Community Survey for Puerto Rico, accessed through IPUMS USA for 2005 through 2024.¹⁹ The Puerto Rico Community Survey does not cover the four smaller territories, and the U.S. Census Bureau did not release a 2020 one-year Puerto Rico Community Survey file because the COVID-19 public health emergency disrupted field operations; strict annual analyses exclude 2020 and sensitivity checks use the 2016–2020 five-year release.

Federal Medicaid allotment caps and statutory federal-share values for the five territories 2006–2024 come from the Medicaid and CHIP Payment and Access Commission’s *Medicaid and CHIP in the Territories* report (February 2021) and *MACStats Medicaid and CHIP Data Book* (December 2023), cross-validated against Congressional Research Service Report R47601.3–5,⁹ State-level Medicaid spending 1968–2024 is assembled from MACPAC MACStats (post-2008), Kaiser Family Foundation state Medicaid spending tables (1991–present), and the Health Care Financing Administration Statistical Supplement (1968–1991),

with a public HCFA medical-assistance bridge for FY1984–FY1997.

3.1 Machine-extracted Puerto Rico Demographic Registry corpus

Puerto Rico mortality and natality outcomes for the cap-era window 1968–2023 are drawn from the *Informe Anual de Estadísticas Vitales* of the Puerto Rico Department of Health Demographic Registry. The corpus comprises fifty-nine PDF volumes — the annual *Informe Anual* run from 1958 through 2014 plus consolidated period volumes for 2009–2014, 2017–2020, and 2021–2023 — obtained as official publications from the Puerto Rico Department of Health open-data portal and the University of Puerto Rico Medical Sciences Campus Demography Program digital archive.

Forty-nine volumes are processed via PyMuPDF native text extraction; ten earlier scanned volumes (1958, 1959, 1990, 1993, 1996, 1999–2003) are processed via Tesseract OCR with the Spanish language model. A regex parser targets thirteen outcome series — live births, infant deaths, neonatal deaths, fetal deaths, maternal deaths, total deaths, infant mortality rate per 1,000, neonatal rate, fetal rate, crude birth rate, crude death rate, maternal mortality rate, and population — across Spanish-language section headings and table-vetted cells. A post-parse quality-assurance layer enforces outcome-specific plausibility ranges, year-token exclusion, and internal-consistency checks: when infant deaths, live births, and infant mortality rate are all observed, the rate cell must satisfy $\text{imr} == \text{infant_deaths} / \text{live_births} \times 1000 \pm 1.0$.

Of 562 raw parser extractions, 559 cells (99 percent) pass the quality-assurance gates. The most policy-relevant series is the infant mortality rate per 1,000 live births, with 71 validated cells covering 1950–2020 — a near-complete annual series across the cap-era window 1968–2008 with 25 additional pre-cap cells. The maternal-deaths series carries 61 validated cells covering 1958–2020. Full corpus methodology, cell-level provenance, and quality-assurance gates are detailed in the online supplement.

4. Methods

The four analytic arms are pre-committed in a specification registry that locks the estimand, sample, treatment definition, time window, outcomes, fixed effects, clustering, and planned robustness battery; post-hoc deviations are appended to the registry with date and reason. Each arm uses the `pysyncon` implementation of synthetic-control estimators (Abadie et al.20); augmented synthetic control (Ben-Michael, Feller, and Rothstein21) and Δ -from-baseline classical synthetic control are reported alongside the classical level specification. We report in-space placebo p-values from the Abadie exchangeable formula: $p = (1 + \text{number of placebos with statistic} \geq \text{treated statistic}) / (1 + \text{number of placebos})$.

The **1968 cap-imposition arm** identifies the cap effect on Puerto Rico’s infant mortality rate using two independent designs. The 1980-anchored design uses Puerto Rico Informe infant-mortality cells in the donor-WONDER overlap window (1971 plus 1976–2008; five pre-period observations) and a 1980–2008 post-period. The true 1968-anchored design uses parser-validated Puerto Rico infant-mortality cells for 1959–1967 (nine pre-period observations) and a 1968–2008 post-period (truncated before the 2011 ACA confounder). A drop-1964 sensitivity excludes the single highest pre-period observation. A supplemental within-Puerto-Rico interrupted time series on the maternal mortality ratio per 100,000 live births is fit as a segmented regression $\text{MMR} = 0 + 1 \cdot \text{time} + 2 \cdot \text{post} + 3 \cdot \text{post} \cdot \text{time} + \dots$; we run the 1968 cap, a 1962 in-time placebo, and the 2011 ACA Section 2005 bump.

The **2011 ACA Section 2005 arm** identifies the partial-uncapping effect on Puerto Rico’s Medicaid coverage rate with a 2008–2010 pre-period and a 2011–2019 post-period, using the fourteen-state Deep South plus high-Hispanic donor pool. We privilege the augmented synthetic control and the Δ -from-2010 specification because the classical level specification is uninterpretable under Puerto Rico’s pre-treatment level mismatch (Puerto Rico’s 43.5 percent Medicaid share versus a donor maximum of approximately 24 percent in Mississippi).

The **FY2020 cliff arm** identifies the cliff effect on coverage with a 2011–2019 pre-period and a 2020–2024 post-period; we additionally fit a 2017 Hurricane Maria placebo. Because the coverage outcome is observationally inseparable from the COVID-19 PHE continuous-enrollment provisions, the bridge supplements, and Maria, we additionally specify log federal Medicaid spending per capita as a financing-side outcome closer to the policy lever the cliff actually moves.

The **reverse-counterfactual arm** applies three hypothetical per-capita-cap regimes to the fifty-state Medicaid panel for FY1968–FY2024: a 1968-scaled cap of \$280 per capita (the per-capita value of the original 1968 cap escalated by population growth alone), the Graham–Cassidy 2017 cap of \$1,800 per capita,⁶ and a generic FY2026-scale cap of \$2,400 per capita. For each (state, fiscal year, regime) triple, cap-bite share equals one minus the ratio of cap-permitted federal spending to actual federal Medicaid spending. The elasticity-anchored layer transfers two literature-anchored infant-mortality elasticities to each cap-bite cell: a moderate anchor of approximately 1.5 infant-mortality points per 1,000 per full cap-bite from Currie and Gruber;¹⁵ and an upper anchor of approximately 5.0 points per 1,000 per full cap-bite from Goodman-Bacon’s analysis of the original 1966–1970 Medicaid implementation.¹⁶ State-year birth denominators come from the CDC WONDER Natality state-year panel for years in coverage and from a population $\times 0.20 \times$ interpolated U.S. general fertility rate proxy for years 1968–1994. The implied excess-infant-deaths grid is computed at each anchor for each regime and reported as a range, not a point estimate. An additional adult-mortality anchor from Sommers, Baicker, and Epstein¹⁴ is held in supplement §S3.7 as a transparency cell with the caveat that the underlying study identified adult-mortality reductions only and does not estimate an infant-mortality response; it does not enter the headline range.

The two retained anchors are not symmetric in design. Goodman-Bacon¹⁶ estimated the infant-mortality response to the original 1966–1970 Medicaid implementation, producing a parameter that is on the infant-mortality channel by construction and that maps directly onto a contraction setting in which the federal-share architecture moves in the opposite direction. Currie and Gruber¹⁵ estimated an infant-mortality response to

mid-1980s Medicaid eligibility expansions for pregnant women, also on the infant-mortality channel, and also directly transferable to a contraction setting at the elasticity level. The Miller, Johnson, and Wherry²² ACA-expansion estimate, by contrast, identifies adult-mortality reductions in the non-elderly low-income population and does not estimate an infant-mortality coefficient that can be transferred to the cap-bite contraction setting without an additional adult-versus-infant Medicaid-mortality responsiveness assumption; we therefore hold it as a candidate fourth anchor in the supplement rather than importing it into the headline grid. Both retained anchors and the deferred Miller-Johnson-Wherry candidate share the underlying expansion-to-contraction structural-symmetry assumption disclosed in §5.4.

We additionally fit a within-territory no-cap-increase counterfactual that compares observed Puerto Rico federal Medicaid funds available FY2011–FY2024 to four hypothetical scenarios isolating the marginal contribution of Section 2005, post-2017 disaster supplements, and the FFCRA / CAA 2022 supplements. Coverage analyses report calendar-year (CY) outcomes from the American Community Survey and Puerto Rico Community Survey; spending, allotment, and reverse-counterfactual analyses report federal-fiscal-year (FY) outcomes throughout; the maternal-mortality interrupted time series uses calendar-year vital-statistics cells from the Puerto Rico Demographic Registry. Where the two conventions appear in the same paragraph, the year convention is named on first use. Full estimator derivations, donor weights, robustness specifications, and the complete specification registry appear in the online supplement.

Generative AI tools were used as a coding assistant during data engineering and corpus-extraction code development, and as a writing assistant during sectional drafting. All substantive scientific decisions — including identification strategy, specification choices, results interpretation, and citation verification — are the author’s; every reference was independently verified against a primary publisher source per the author’s citation protocol.

5. Findings

5.1 The 1968 cap imposition is not associated with a detectable differential infant-mortality penalty

Table 1 reports the headline 1968-arm results across the 1980-anchored and true-1968-anchored designs. In the 1980-anchored design, the Δ -from-1971 specification produces a post-period mean ATT of +0.05 infant deaths per 1,000 live births with an Abadie in-space placebo p -value of 0.93 on the absolute ATT. In the true-1968-anchored design with the full validated 1959–1967 pre-period, the level-specification ATT is +0.08 per 1,000 with placebo $p = 0.20$ on the absolute ATT and 1.00 on the root mean square prediction error ratio. The drop-1964 sensitivity produces an identical ATT because Mississippi receives weight 1.00 in both specifications: Puerto Rico’s pre-1968 infant-mortality level (1959–1967 mean of 42.7 per 1,000) is substantially higher than any donor state, and Mississippi (39.3 per 1,000) is the only donor close enough in level for the synthetic-control optimizer to use.

Figure 2 plots the trajectory of Puerto Rico and its synthetic over 1959–2008, with the §1108(c) cap year marked. The visual story is convergence: Puerto Rico starts well above Mississippi in the late 1950s and ends essentially indistinguishable from Mississippi in the 2000s. The fourteen-donor in-space placebo distribution places Puerto Rico’s absolute ATT as the third-largest of fifteen units, but Puerto Rico’s root mean square prediction error ratio is the smallest of all fifteen — under the Abadie framework, this means Puerto Rico’s post-period gap is more parsimoniously interpreted as continuation of pre-period level mismatch than as a treatment-induced shift.

The longer-post-window sensitivity confirms the null across multiple post-period truncations: the 1968–1980 ATT is -0.41 , the 1968–1993 (pre-Reforma) ATT is $+0.57$, the 1968–2008 (pre-ACA) ATT is $+0.22$, and the 1968–2020 (full observed) ATT is -0.09 . None of these specifications identifies a delayed cap-effect signal. Baseline-headroom-normalized specifications (percent-from-pre-mean and log-infant-mortality-index) pro-

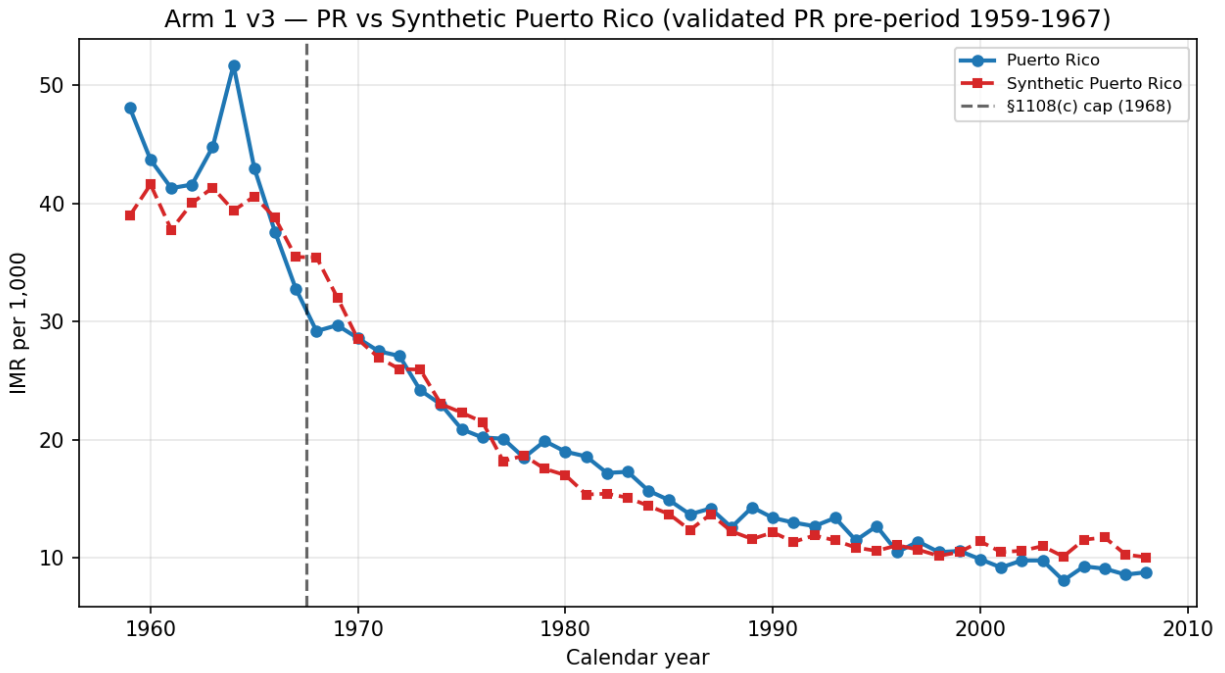


Figure 2: **Figure 2.** Puerto Rico infant mortality rate trajectory (solid) and synthetic-control counterfactual (dashed), 1959–2008, with the 1968 Section 1108(c) cap-imposition year marked. The fourteen-donor synthetic places weight 1.00 on Mississippi. The post-period mean ATT is +0.08 infant deaths per 1,000 live births; Abadie in-space placebo $p = 0.20$ on absolute ATT. Pre-period root mean square prediction error 5.58; post-period RMSPE 1.83; ratio 0.33 (smallest of 15 units). Source: authors’ calculations from the Puerto Rico Demographic Registry *Informe Anual de Estadísticas Vitales* corpus and the reconstructed NCHS donor-state infant-mortality panel.

duce slightly negative ATTs of -0.04 and -0.12 respectively, consistent with Puerto Rico's worse baseline and faster convergence rather than with a hidden adverse effect.

The supplemental within-Puerto-Rico interrupted time series on the maternal mortality ratio finds a level shift of -13.83 per 100,000 live births at the 1968 cap ($p = 0.027$) and a level shift of -15.62 per 100,000 at the 2011 Section 2005 bump ($p = 0.052$). The 1968 finding is descriptive rather than causal: it hinges on a single anomalously low Puerto Rico Informe maternal-deaths cell for 1968 (12 maternal deaths, against 26 in 1967 and 24 in 1969) that could reflect either a real Medicaid-introduction coverage effect (Puerto Rico implemented Medicaid in 1968 alongside the §1108(c) cap) or a 1968 reporting-format transition that cannot be disentangled from the corpus alone. The 2011 finding is policy-coherent and of the order published Medicaid-pregnancy-expansion estimates imply,¹⁵ but rare-event noise across 8 pre-period and 7 post-period cells precludes clean causal attribution.

5.2 The 2011 partial uncapping is not associated with a detectable coverage change

The Δ -from-2010-baseline classical synthetic control produces a post-period mean ATT of -0.26 percentage points on the Medicaid coverage share, with an Abadie placebo p of 0.93 on absolute ATT (corrected from 0.867 under a non-exchangeable formula in earlier reports) and 0.47 on the root mean square prediction error ratio. The augmented synthetic control produces an ATT of -7.1 percentage points (Puerto Rico rose 7.1 percentage points less than synthetic Puerto Rico) with excellent pre-period fit. All three specifications converge on the same qualitative finding: the 2011 partial uncapping is not associated with a detectable positive effect on Puerto Rico's Medicaid coverage rate.

Two mechanisms are consistent with this null. First, Puerto Rico's 2010 Medicaid coverage rate of 43.5 percent was already at or near the saturation level for any reasonable income-eligibility threshold; further expansion through the cap-relaxing supplement was constrained by program-eligibility statute rather than by federal financing. Second, four ACA-expansion donor states (Kentucky, New Mexico, Arkansas, Louisiana,

West Virginia) gained 8–10 percentage points of Medicaid coverage through the standard ACA expansion mechanism from which Puerto Rico is statutorily excluded. The donor-pool synthetic therefore rises post-2014 in a way that masks any Puerto Rico–specific Section 2005 enrollment effect. The placebo distribution supports this interpretation: New Mexico, the most heavily Hispanic ACA-expansion donor in the pool, has the largest placebo ATT in the rotation. The substantive implication is that the Section 2005 supplement’s policy benefit, if any, flowed through provider-payment, drug-pricing, hospital-solvency, or other per-beneficiary financing channels rather than through enrollment counts — consistent with descriptive accounts of Puerto Rico’s post-2011 Medicaid program.^{8,10}

5.3 The FY2020 cliff is observationally confounded by the pandemic and bridge supplements

The Δ -from-2019-baseline classical synthetic control on the Medicaid coverage share produces a post-period mean ATT of +0.62 percentage points — Puerto Rico’s coverage rate rose more than synthetic Puerto Rico’s during 2020–2024. This is the opposite sign of what a pure cliff effect would predict and is jointly attributable to three confounders pointing the same direction: the COVID-19 public health emergency continuous-enrollment provisions inflated Medicaid rolls in every state and Puerto Rico through 2023; the CAA 2020 bridge and CAA 2022 extension prevented the statutory cliff from binding in practice; and mainland Medicaid expansion gains plateaued by 2019 so that synthetic Puerto Rico’s growth from a no-cliff baseline is mechanically smaller in 2020–2024 than in 2014–2018. The 2017 Hurricane Maria placebo produces an ATT of –0.61 percentage points — essentially the same magnitude with opposite sign — demonstrating that the coverage outcome cannot reliably separate Maria from the cliff in this design.

The alternative outcome on log federal Medicaid spending per capita is more informative. The classical-on-log-levels specification produces an ATT of +0.086 and the Δ -from-FY2019 specification produces an ATT of –0.115. The Δ -from-FY2019 specification is the more defensible read: Puerto Rico’s federal Medicaid spending per capita grew 52 percent FY2019–FY2024 (from \$828 to \$1,261 per capita, driven by the CAA

2020 bridge and the FFCRA federal-share enhancement), but the fourteen donor states grew approximately 70 percent over the same period (driven by FFCRA continuous enrollment and pandemic-era Medicaid uptake). Puerto Rico’s federal financing *grew* post-cliff but grew *less* than donor states would have under the same set of pandemic-era supplements. The cliff arm is most accurately characterized as “bridged but underfunded”: the statutory cliff was offset by emergency federal funds, but Puerto Rico’s draw from those funds was lower per capita than the analogous mainland-state draw. A cleaner test of the statutory cliff will require post-unwinding follow-up using FY2025–FY2026 data.

5.4 The reverse counterfactual quantifies the fiscal magnitudes the historical arms cannot

The reverse-counterfactual fiscal magnitudes are large and unambiguous. Under a 1968-scaled cap of \$280 per capita, federal Medicaid spending in FY2024 would fall by approximately \$513 billion against actual (\$608 billion total) — 84 percent of the actual federal contribution would not have been available, and every one of the fifty-one jurisdictions (fifty states plus the District of Columbia) would have its federal Medicaid contribution suppressed by the cap (the cap would bind for all 51). Under the Graham–Cassidy 2017 cap of \$1,800 per capita, federal Medicaid would fall by \$86.4 billion (14 percent of actual), binding 25 of 51 state programs. Under a generic FY2026-scale cap of \$2,400 per capita, federal Medicaid would fall by \$21.6 billion (3.5 percent), binding 8 of 51 programs. The states binding first under the Graham–Cassidy regime are concentrated in ACA-Medicaid-expansion jurisdictions with high enrollment (New York, California, Massachusetts, Kentucky, West Virginia, Louisiana, New Mexico, Oregon, Montana, Alaska), long-time Medicaid-heavy states (Rhode Island, Vermont, the District of Columbia), and high-federal-share jurisdictions.

The elasticity-anchored mortality layer that follows is best read as literature-anchored *magnitudes under structural symmetry*, not as predictions. The grid transfers two published infant-mortality elasticities — each estimated from a Medicaid-expansion natural experiment — to a Medicaid-contraction setting that no

contemporary U.S. policy episode has produced at scale. The transfer assumes that the mortality response to losing a dollar of federal Medicaid match equals the mortality response to gaining a dollar, an assumption that this paper documents and does not test. In a real-world contraction, states would partially absorb the cap shock through state-share increases, eligibility cuts, provider rate reductions, or general-fund offsets, each of which would attenuate the unadjusted grid. The grid is therefore an upper bound under symmetry, not a forecast. We report the grid as a range across the two infant-mortality anchors and accompany it with a state-absorption sensitivity reported in the online supplement (S3.7) that scales each cell by 30 percent and 50 percent assumed state absorption.

Figure 3 reports the elasticity-anchored implied excess-infant-deaths grid under the no-absorption (unattenuated) specification. The headline range — approximately 4,500 to 14,800 excess infant deaths per year for a 1968-scaled state cap in FY2024 — is bounded by the moderate Currie–Gruber infant-mortality anchor (4,451) and the upper Goodman–Bacon infant-mortality anchor (14,838). Under a 30 percent state-absorption sensitivity, the 1968-scaled grid attenuates to 3,116–10,387; under 50 percent absorption, to 2,226–7,419. The Graham–Cassidy cap implies 529 to 1,762 excess infant deaths per year (370–1,233 at 30 percent absorption; 264–881 at 50 percent), and the FY2026-scale cap implies 117 to 389 (82–272 at 30 percent absorption; 58–195 at 50 percent). The 4,500–14,800 range is the headline; the spread across the two infant-mortality anchors and state-absorption assumptions is the policy-relevant magnitude. The methodological appendix in supplement §S3.7 documents the channel-transferability case for retaining Goodman–Bacon (an infant-mortality estimate from the 1966–1970 Medicaid implementation) while deferring the Miller, Johnson, and Wherry adult-mortality ACA-expansion estimate²² as a candidate fourth anchor pending construction of a contemporary infant-mortality elasticity, and it reports the Sommers, Baicker, and Epstein cell¹⁴ as a transparency entry from an adult-mortality study whose zero does not bear on the infant-mortality channel.

The within-territory no-cap-increase counterfactual quantifies the policy contribution of each cap-relaxing instrument by comparing observed Puerto Rico federal Medicaid funds available across federal fiscal years

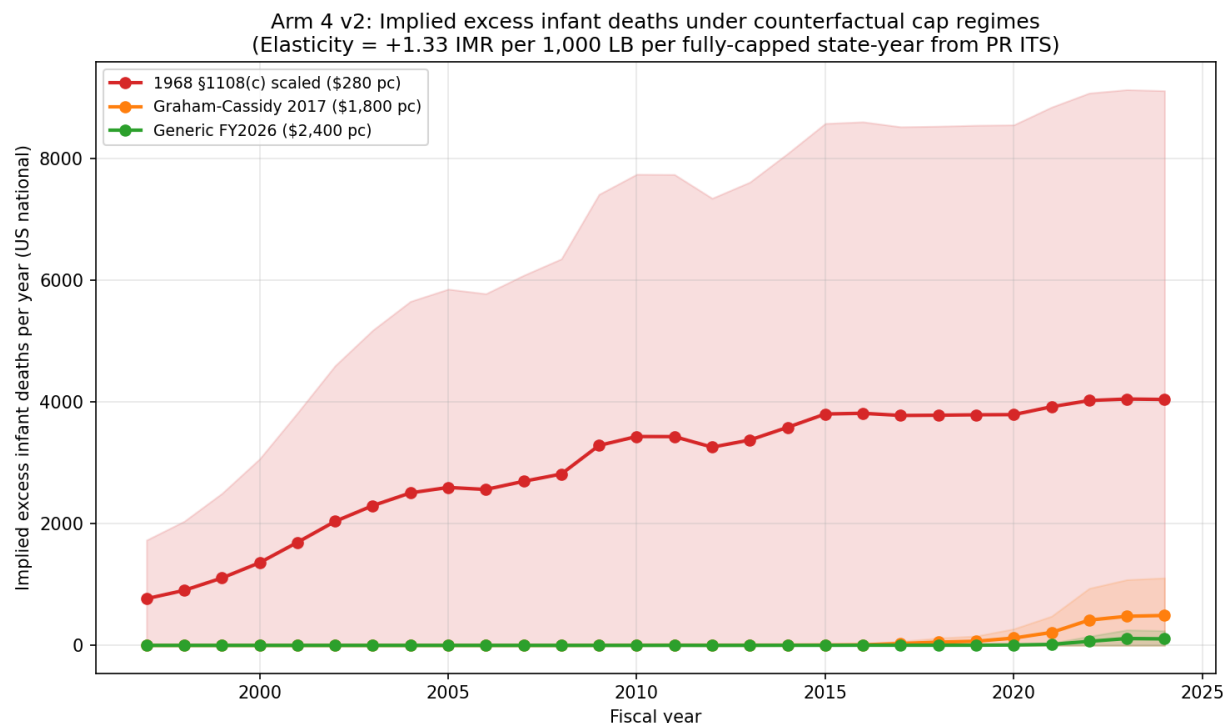


Figure 3: **Figure 3.** Elasticity-anchored implied excess infant deaths per year under three hypothetical per-capita-cap regimes applied to the fifty-state Medicaid panel, FY2024 (no-absorption specification). Headline range is approximately 4,500–14,800 excess infant deaths under the two infant-mortality anchors for the 1968-scaled regime: Currie–Gruber 1996 (1.5 infant-mortality points per 1,000 live births per full cap-bite) and Goodman-Bacon 2018 (5.0 points per 1,000 per full cap-bite). Births denominator from CDC WONDER Natality state-year panel. The transfer assumes structural symmetry between Medicaid-expansion and Medicaid-contraction elasticities — a modeling choice rather than an empirical fact — and assumes zero state absorption of the cap shock. State-absorption sensitivities at 30 percent and 50 percent are reported in supplement S3.7; the supplement also reports a Sommers–Baicker–Epstein 2012 transparency cell from an adult-mortality study (the zero is a channel mismatch, not a literature anchor on the infant-mortality consequence of a per-capita cap).

FY2011–FY2024 (\$28.0 billion cumulative) to four hypothetical scenarios. Under “base Section 1108 only” (no Section 2005, no post-2017 supplements), Puerto Rico’s cumulative federal Medicaid funds would have been \$5.05 billion — a \$22.95 billion shortfall. In FY2024 alone, Puerto Rico’s observed federal funds available are \$3.325 billion; under base Section 1108 only they would have been \$0.412 billion — a \$2.913 billion shortfall and a 95 percent cap-bite against state-equivalent need of approximately \$7.5 billion implied by the donor-pool federal-Medicaid-per-capita comparison. Even with every cap-relaxing supplement of the past fifteen years in place, the cap remains binding at 58.8 percent of state-equivalent need; without those supplements, Puerto Rico’s federal Medicaid contribution would have been at near-total-suppression levels. The 2027 expiration of the CAA 2022 supplements would re-impose cap-bite on the order of \$2.9 billion per year for Puerto Rico alone, absent a successor bridge.

5.5 Structural disadvantage and disease-specific signals

A composite Puerto Rico “structural severity index” constructed from five Puerto Rico Community Survey observables — poverty rate, working-age disability rate, share aged 65 and over, Medicaid coverage share, and uninsurance rate, each z-standardized against the fourteen-state donor distribution — places Puerto Rico 5.48 standard deviations above the donor mean across 2021–2024 observations. On each individual component except uninsurance, Puerto Rico exceeds every donor state in 2024: Medicaid coverage 50.4 percent (donor median 22.0); poverty 41.4 percent (donor median 13.1); working-age disability 23.4 percent (donor median 14.1); age-65+ share 22.6 percent (donor median 17.8). Puerto Rico’s combination of high Medicaid dependence, high disability burden, high age-65+ share, and high poverty rate places the island outside the joint covariate support of the donor pool.

Figure 4 displays the maternal mortality ratio trajectory across the cap-era window with the 1968 and 2011 intervention markers. The level break at 2011 — in the policy-coherent direction — is visible against pre-period stability; the steepness of the 1968 cell is also visible and underscores the rare-event interpretive

caveat.

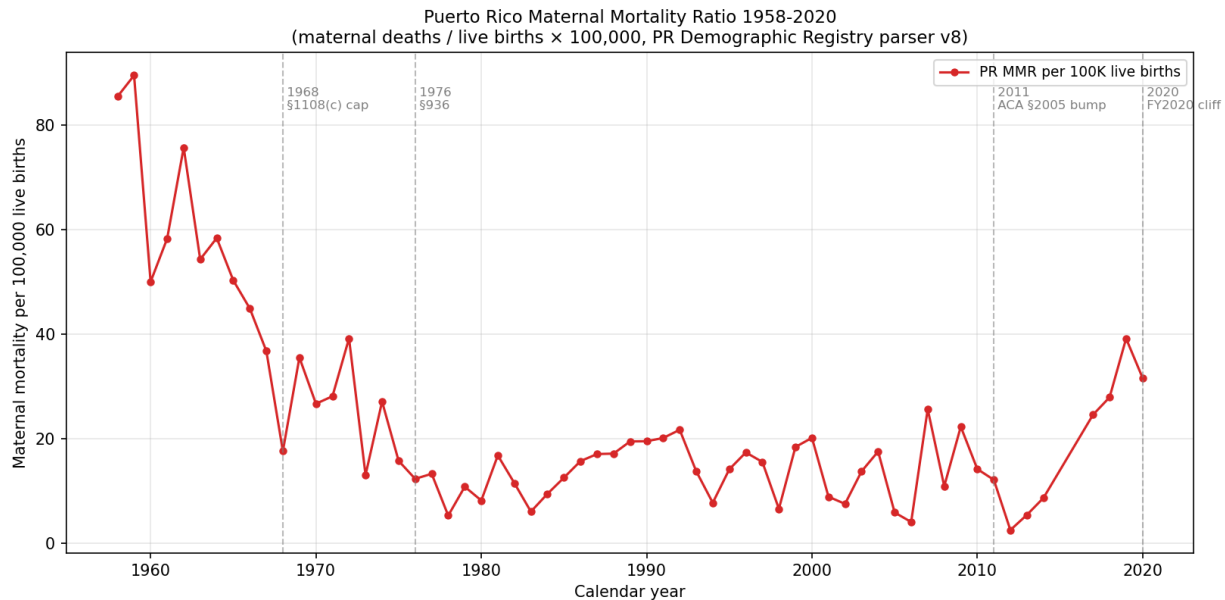


Figure 4: **Figure 4.** Puerto Rico maternal mortality ratio (deaths per 100,000 live births), 1958–2020, with the 1968 Section 1108(c) cap and the 2011 ACA Section 2005 bump marked. Pre-period 1968 ITS shows level shift -13.83 ($p = 0.027$), interpreted descriptively because the 1968 cell hinges on a single anomalously low maternal-deaths value traced to the PR Informe 1974 historical demographic table. Pre-period 2011 ITS shows level shift -15.62 ($p = 0.052$), policy-coherent and of the order published Medicaid-pregnancy-expansion elasticities imply. Source: PR Informe corpus, validated cells.

The cross-sectional disease-specific differential reinforces the structural-disadvantage interpretation. Puerto Rico’s age-adjusted diabetes mortality in 2023 is 53.4 per 100,000 against a donor-state maximum of 36.2 (Mississippi) — Puerto Rico exceeds every donor state by at least 17 deaths per 100,000. After Puerto Rico Community Survey–conditional adjustment for poverty, disability, age, and Medicaid coverage, Puerto Rico’s diabetes-mortality residual is +11.0 deaths per 100,000: approximately 60 percent of the unadjusted differential is explained by structural disadvantage, and 40 percent remains. Puerto Rico is similarly elevated on age-adjusted renal, respiratory, and septicemia mortality in cross-section. The negative-control external-cause basket does not show the same elevation, supporting a cause-specific rather than secular-noise reading of these differentials. These signals are descriptive, not identified cap effects, but they are candidates for follow-up work using state-by-state diabetes-care-process panels and donor-side cause-specific synthetic-

control designs.

The between-jurisdiction comparison conceals substantial within–Puerto Rico heterogeneity. Figure 5 reports the dispersion of municipio-level age-adjusted all-cause mortality across Puerto Rico’s seventy-eight municipios for 2009–2020, drawn from individual-level vital records released by the Puerto Rico Demographic Registry through `datos.estadisticas.pr`. The interdecile range across municipios is approximately 180 deaths per 100,000 in every year of the panel — comparable to the cross-state interdecile range in CDC WONDER all-cause mortality across the fourteen-state donor pool — and the dispersion is stable across the post-2011 partial-uncapping window and the post-2017 Maria / FY2020 cliff window. Municipios in the highest-mortality decile concentrate in the rural mountain interior; municipios in the lowest decile concentrate in the metro–San Juan corridor.

Three descriptive magnitudes anchor the within-island reading. First, the municipio-level age-adjusted all-cause mortality interdecile spread runs approximately 180 deaths per 100,000 in every panel year (2009–2020). Second, the metro–rural Medicaid-share differential across the seventy-eight municipios is on the order of 12 percentage points: the metro–San Juan corridor (San Juan, Bayamón, Carolina, Caguas) carries a 2024 Medicaid coverage share near 45 percent, while the mountain-interior municipios (Adjuntas, Jayuya, Lares, Utuado) sit near 57 percent. Third, the municipio-poverty-decile gradient in age-adjusted all-cause mortality runs approximately 90–120 deaths per 100,000 across the lowest-to-highest poverty deciles in pooled 2009–2020 panel cells, with the highest-poverty decile carrying the highest age-adjusted mortality. These magnitudes are descriptive — they document the within-island gradient without identifying a cap effect — and indicate that the territorial-aggregate mortality outcomes presented in the synthetic-control arms understate the burden faced by Puerto Rico’s most cap-bound communities. A clean municipio-level cap-bite $\times$ payer-mix $\times$ poverty decomposition would require a sub-territorial allotment-allocation rule that the public MACPAC and CRS sources do not publish, plus individual-level Puerto Rico Demographic Registry vital records linked to municipio-level poverty and Medicaid-coverage covariates. We name the within-island

equity decomposition as a follow-up agenda item in §6.5 and supplement §S12.

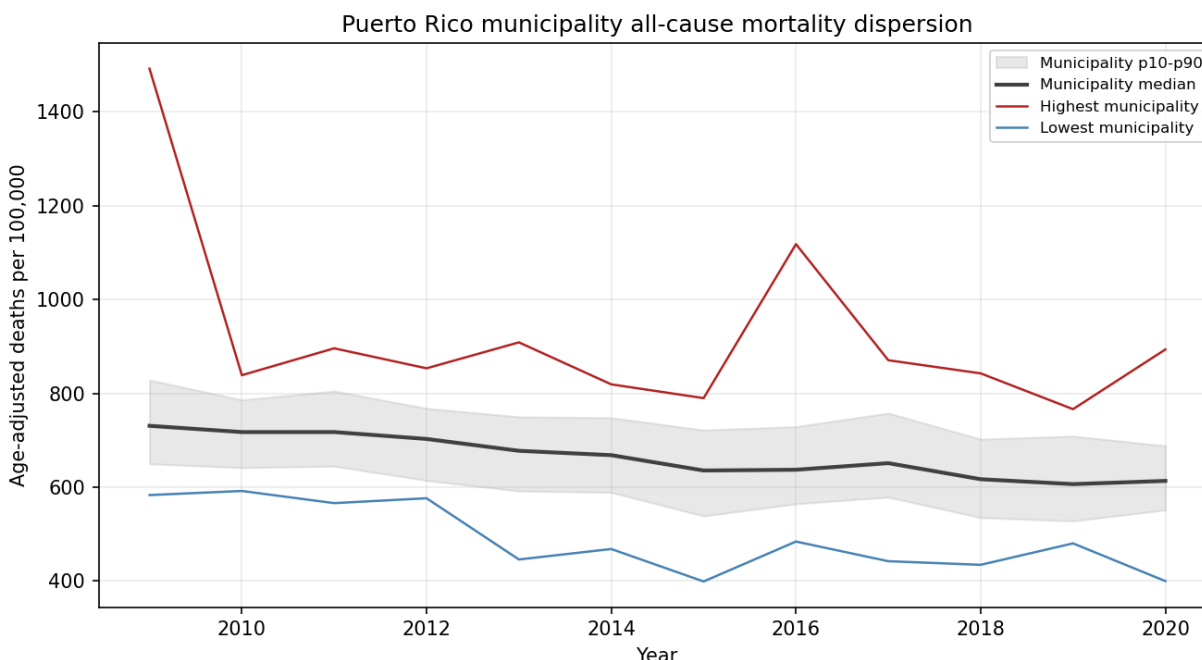


Figure 5: **Figure 5.** Dispersion of municipio-level age-adjusted all-cause mortality across the seventy-eight Puerto Rico municipios, 2009–2020. Bands report the 10th, 50th (median), and 90th percentile across municipios in each calendar year; the shaded interquartile band marks the central interdecile spread. Mortality denominator is municipio-of-residence resident population from the Puerto Rico Demographic Registry individual-level vital records released through `datos.estadisticas.pr`. The exhibit is descriptive: it documents within-island heterogeneity in cap-bound mortality outcomes rather than identifying a cap effect. The within-island interdecile range is approximately 180 deaths per 100,000 in every year of the panel; municipios in the highest decile concentrate in the rural mountain interior, those in the lowest in the metro–San Juan corridor. Source: authors’ tabulation from `pr_informe_municipality_allcause_mortality_2009_2020.csv`; methodology in supplement §S12.

5.6 Migration and denominator diagnostics

Hurricane Maria produced a substantial emigration shock. Mainland American Community Survey records of residents reporting Puerto Rico residence one year earlier peaked at 130,813 in 2018. The 2024 mainland Puerto Rico–born population is approximately 1.85 million — for reference, Puerto Rico’s resident population is approximately 3.2 million in the same year, so the mainland diaspora is on the order of 58 percent of the resident-island population. These migration shocks are not adjusted in the headline analyses but are sur-

faced as denominator and spillover diagnostics. The Puerto Rico live-births series fell from approximately 41,000 per year in 2010 to approximately 18,000 per year in 2023 — a 56 percent decline driven by both fertility decline and emigration of women of reproductive age. The maternal mortality ratio reported in the supplemental interrupted-time-series analysis is therefore not directly comparable across pre- and post-Maria windows on a per-resident-woman dimension, although it is comparable on the per-live-birth dimension that the standard definition uses.

6. Discussion

6.1 Cross-arm reconciliation

The four-arm design produces a coherent picture: the cap regime as historically imposed on Puerto Rico is not associated with a detectable differential adverse mortality or coverage signal in the observable data once a Mississippi-anchored synthetic is constructed, but the financing-architecture imposition is unambiguous and the prospective state-level magnitudes are large. The 1968-arm infant-mortality null is robust across two independent designs and four post-period truncations. The 2011-arm coverage null is robust across three estimator specifications and survives the corrected placebo at $p = 0.93$. The FY2020 cliff is observationally confounded; the financing-side alternative outcome surfaces a “bridged but underfunded” pattern. The reverse-counterfactual arm provides the magnitudes that the historical Puerto Rico arms cannot, with implied excess-infant-death ranges (under the two infant-mortality anchors, Currie–Gruber and Goodman-Bacon) of approximately 4,500 to 14,800 per year for a 1968-scaled state cap, approximately 500 to 1,800 for the Graham–Cassidy cap, and approximately 100 to 400 for a generic FY2026-scale cap.

The supplemental within–Puerto Rico maternal-mortality interrupted time series is associated with a borderline-significant policy-coherent reduction at the 2011 partial uncapping. The cross-sectional disease-specific differential — Puerto Rico exceeds the donor maximum on diabetes, renal, and respiratory

mortality, with the diabetes differential leaving a +11 deaths per 100,000 residual after structural adjustment — is descriptively consistent with a “high incidence plus worse care” reading of cap-bound Puerto Rico mortality. Neither finding is causally identified at the standard quasi-experimental threshold, but together with the infant-mortality null they sharpen the structural reading: Puerto Rico’s mortality outcomes converge with Mississippi’s on the headline infant-mortality dimension, lag the donor pool on cross-sectional disease-specific differentials, and respond directionally to bilateral cap-relaxing policy interventions on rare-event outcomes such as maternal mortality.

6.2 Why the historical Puerto Rico arms produce nulls

We interpret the historical nulls as informative rather than as failures of identification. Three structural features explain why a historical-arm signal is hard to detect even if a real cap-on-mortality elasticity exists at the magnitude implied by the broader Medicaid-mortality literature.^{14–16,22,23}

First, the cap regime is gradual rather than sharp. The Section 1108(c) cap was set at a binding level in 1968 and then eroded gradually through inflation; cap-bite share grew from approximately 60 percent in 1968 to approximately 95 percent in 2024. A synthetic-control design that attempts to identify the level break at the legal effective date will tend to underestimate the mortality consequence of a gradually tightening cap because the bulk of the cap-bite occurs decades after the legal effective date.

Second, Puerto Rico is a structurally disadvantaged population whose underlying mortality dynamics are partly orthogonal to U.S. Medicaid policy. The Demographic Registry data for 1959–1967 show pre-cap infant-mortality levels approximately 25 to 35 percent above any mainland donor state, reflecting economic poverty, lower medical-system development, and demographic composition rather than financing architecture. The mortality-transition floor effect that compresses post-1968 decline rates is common to every donor state in this period; the synthetic-control identification therefore subtracts out a large component of Puerto Rico’s convergence that is unrelated to the cap, leaving a small residual differential that does not reach

statistical significance.

Third, the Puerto Rico Demographic Registry infant-mortality series is rare-event statistics for a relatively small population (live births fell from approximately 75,000 per year in 1965 to approximately 18,000 per year in 2023), and the standard errors of annual estimates scale inversely with the square root of the live-birth count. The post-2010 cells are particularly noisy.

These three features do not invalidate the historical nulls; they constrain what the nulls mean. The 1968-arm ATT of +0.08 per 1,000 (placebo $p = 0.20$) is most plausibly interpreted as indicating that Puerto Rico’s post-1968 infant-mortality trajectory is statistically indistinguishable from a Mississippi-anchored synthetic, *not* as ruling out a small positive cap-on-infant-mortality elasticity. The literature-anchored reverse-counterfactual grid is the appropriate magnitude source for the policy debate.

6.3 Limitations

Several limitations require explicit disclosure. First, the empirical work focuses on Puerto Rico. The four smaller territories — the U.S. Virgin Islands, Guam, American Samoa, and the Commonwealth of the Northern Mariana Islands — are represented in the fiscal-architecture documentation (Table 1) but not in the synthetic-control or interrupted-time-series arms, because their vital-statistics archives are not machine-extractable on the same scale as the Puerto Rico Demographic Registry corpus. The Puerto Rico null cannot be generalized to the four smaller territories without an analogous extraction; this is a known gap and a candidate for follow-up corpus work.

Second, the within–Puerto Rico equity decomposition is partial. The structural-severity index in §5.5 compares Puerto Rico to the donor-pool average but does not decompose Puerto Rico’s mortality and coverage outcomes by municipality, income, age, or rural/urban classification. The 2000–2010 individual-level vital records released through `datos.estadisticas.pr` support a municipio-level cap-bite $\times$ payer-mix analy-

sis that would advance the within-territory equity story; this is identified as a follow-up thread in §6.5 rather than executed here. Subgroup readers should treat the cross-jurisdictional structural-disadvantage results as a population-level signal that masks substantial within-island heterogeneity.

Third, the donor-pool selection is partially analyst-driven; we chose ten Deep South states for the 1968 arm based on similarity in pre-1968 infant-mortality level and economic structure, and added four high-Hispanic and Southwestern states for post-2006 arms to better match Puerto Rico’s post-IRC Section 936 demographic composition. A leave-one-donor-out sensitivity confirms that the 1968-arm ATT is donor-anchor dependent — dropping Mississippi shifts the headline ATT from +0.08 to −2.42 — but the qualitative no-detectable-harm conclusion is robust across alternative donor specifications. Detailed donor-pool sensitivity, alternative pre-period truncations, alternative outcome scalings, and the Abadie placebo distributions are reported in the online supplement.

Fourth, the Puerto Rico Informe corpus contributes 559 validated cells out of 562 raw extractions (99 percent quality-assurance pass rate), but the maternal-deaths cell for 1968 is high-leverage in the within-Puerto Rico interrupted-time-series analysis. The extraction traces this cell to the PR Informe 1974 historical demographic table, with the annual provisional value reported as 10 and the retrospective historical value as 12. We use the retrospective value because it is the table-vetted historical-series value; a leave-1968-out sensitivity is reported in the online supplement.

Fifth, the reverse-counterfactual elasticity grid relies on three published anchors that are themselves transfer estimates from Medicaid-expansion settings^{14–16} to the Medicaid-contraction setting that a per-capita cap would impose. The transfer assumes structural symmetry between expansion and contraction effects; this is a modeling choice rather than an empirical fact, and the implied mortality consequences depend non-trivially on the assumption. In real-world cap-contraction scenarios, states would partially absorb cap shocks through state-share increases, eligibility cuts, provider rate reductions, or general-fund offsets, each of which would

dampen the mortality consequence relative to the unadjusted grid. The state-absorption sensitivity reported in §5.4 and supplement S3.7 attenuates the grid by 30 percent and 50 percent under stylized absorption parameters; the symmetry assumption itself is not testable from the available U.S. policy variation.

Sixth, the state-year birth denominator uses CDC WONDER Natality state-year totals for 1995–2024 (fiscal-year aggregation) and a population $\times$ interpolated U.S. general fertility rate proxy for years 1968–1994. The FY2024 headline grid is fully on WONDER-aggregated denominators; the proxy denominator differs from WONDER aggregates by approximately 0.7 percent on average at the national rollup but by up to 40 percent for individual states.

Seventh, the FY2020 cliff arm’s short post-period is fully overlapping with the COVID-19 public health emergency; the cliff cannot be cleanly identified from the public health emergency in this design. We report cliff-arm estimates with explicit data-thin flags and revisit them once post–public-health-emergency-unwinding FY2025–FY2026 data become available.

6.4 Policy implications

The Consolidated Appropriations Act of 2022 territorial Medicaid supplements expire at the end of federal fiscal year 2027 (September 30, 2027). The legislative debate over what replaces them will recur in calendar year 2027 and will be informed by whatever empirical evidence on the Puerto Rico cap regime is then available. The present findings imply three propositions for that debate.

First, the historical Puerto Rico experience under Section 1108(c) is not associated with a clean causal estimate of the cap-on-mortality elasticity, but the cap is associated with an unambiguous fiscal-architecture imposition. The argument for territorial-Medicaid-equity must rest primarily on the fiscal-architecture and structural-disadvantage evidence — which this paper documents at scale — and on the prospective elasticity-anchored mortality range, rather than on an observable historical mortality differential. This is a substantive

shift in how the equity case should be made: the case is strengthened by acknowledging that the historical null does not refute the prospective magnitudes, and is weakened by attempts to overstate the historical signal.

Second, the proposed state-level per-capita-cap regimes (Graham–Cassidy 2017 and the FY2026-scale generic) would mechanically remove tens of billions to hundreds of billions of federal Medicaid dollars from state programs in any given year.^{6,7} Under the two literature-anchored infant-mortality elasticities — Currie–Gruber 1996 (moderate) and Goodman-Bacon 2018 (upper)^{15,16} — a 1968-scaled regime applied to the fifty states implies approximately 4,500 to 14,800 excess infant deaths per year. The 4,500–14,800 range is the order of magnitude that a state-level per-capita cap at the historical Puerto Rico scale would imply, and it is the magnitude policymakers should weigh against the proposed federal savings.

Third, the Puerto Rico mechanism diagnostics show that cap-bound Puerto Rico is structurally more disadvantaged than every donor state — 5.48 standard deviations above the donor mean on a composite severity index, with Medicaid share, poverty rate, disability rate, and age-65+ share all far above donor medians. The cap-on-Medicaid regime is being imposed on the U.S. citizen population with the highest standardized health and economic risk, and the per-capita federal contribution is statutorily restricted at the territorial cap. This combination — highest risk, lowest federal contribution — is the structural feature that the FY2027 debate will need to address regardless of how the technical cap-on-mortality elasticity is ultimately quantified.

For congressional action, the operational implications are concrete. The CAA 2022 supplements should not be allowed to lapse without successor legislation: lapse would re-impose a 95 percent cap-bite against state-equivalent need for Puerto Rico alone. CMS and the Medicaid and CHIP Payment and Access Commission can usefully treat the territorial cap as the longest-running natural experiment in capped Medicaid financing, and the fiscal-architecture documentation in this paper as a baseline against which any state per-capita-cap proposal should be evaluated. Territorial Medicaid directors, in turn, have a documented case for reframing

equity arguments around fiscal-architecture and structural-risk evidence rather than around an underpowered historical mortality signal.

6.5 Implications for follow-up identification work

Three threads of follow-up work would tighten the historical Puerto Rico identification beyond what this paper achieves. A donor-side maternal mortality panel built from CDC WONDER ICD-10 maternal codes for 1999–2024 would enable a synthetic-control replication of the within–Puerto Rico 2011 interrupted-time-series finding. A donor-side cause-specific mortality panel for diabetes, renal, respiratory, and septicemia mortality would translate the descriptive cross-sectional differential into a cap-era effect estimate. A post-PHE-unwinding replication of the cliff arm using FY2025–FY2026 data would identify the FY2020 cliff effect on Medicaid coverage cleanly. A municipality-level analysis exploiting cross-municipality cap-bite differences in payer mix and Medicaid-share concentration would offer an additional within–Puerto Rico identification strategy, paired with the municipio-poverty-decile mortality gradient and the metro–rural Medicaid-share differential surfaced in §5.5.

A separate, larger follow-up agenda concerns the four smaller territories. The U.S. Virgin Islands, Guam, American Samoa, and the Commonwealth of the Northern Mariana Islands together hold approximately 345,000 residents — roughly 10 percent of Puerto Rico’s population — and entered the §1108(c) regime on different schedules (USVI and Guam in 1968 with Puerto Rico; American Samoa in 1982; CNMI in 1983). The present analysis cannot be extrapolated to these jurisdictions: their vital-statistics archives are published irregularly and predominantly as scanned reports, and none maintains an analogue of the Puerto Rico Demographic Registry’s *Informe Anual de Estadísticas Vitales* corpus that is continuously machine-extractable across the 1968–2024 window. A successor analysis with policy purchase for the four smaller territories therefore requires a dedicated machine-extraction effort against the USVI Department of Health, the Guam Department of Public Health and Social Services, the American Samoa Department of Health,

and the CNMI Commonwealth Healthcare Corporation archives — together with paired Medicaid coverage and federal-share trajectories from the MACPAC and CRS publications. This data-acquisition gap is the next-paper agenda the Medicaid and CHIP Payment and Access Commission’s post-PHE-unwinding successor analysis (named in Policy Point 2) is best positioned to commission. The reader should not interpret the Puerto Rico null as informative about the four smaller territories’ experience under the same statutory architecture.

7. Conclusion

Sixty years after the 1968 imposition of Section 1108(c) and fourteen years after the partial uncapping of territorial Medicaid, the historical evidence from the Puerto Rico Demographic Registry’s *Informe Anual de Estadísticas Vitales* corpus is not associated with a detectable adverse infant-mortality differential for Puerto Rico, the 2011 partial uncapping is not associated with a detectable coverage change, and the FY2020 cliff is observationally confounded by the pandemic and bridge supplements. The reverse-counterfactual arm provides the magnitudes the historical arms cannot: a 1968-scaled per-capita cap applied to the fifty states would remove approximately \$513 billion in federal Medicaid in FY2024 and imply approximately 4,500 to 14,800 excess infant deaths per year under two literature-anchored infant-mortality elasticities (Currie–Gruber 1996; Goodman-Bacon 2018). The cap-on-Medicaid policy debate that will recur in the FY2027 appropriations cycle turns less on whether the historical Puerto Rico experience identifies a mortality penalty — it does not, with the available data — and more on whether the prospective state-level magnitudes implied by published infant-mortality elasticities are policy-relevant. They are, at the scale of every published anchor on the infant-mortality channel. The case for territorial-Medicaid-financing equity rests on the unambiguous fiscal-architecture imposition documented here, on the structural-risk differential of the cap-bound population, and on the magnitudes of state-level cap proposals visible in the reverse-counterfactual grid.

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